QE 2025 Final Project Report

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1 Introduction

This document outlines the numerical methods, results, and interpretation of the final project for the Quantitative Economics course. We will discuss:

Numerical Methods

- Grid construction for Captial (k) and Productivity (z)
- Interpolation methods used for value function approximation
- VFI process with golden-section and Howard policy iteration
- MoM Estimation setup for $\{\gamma_{adj}, F, p_s\}$.

Interpretation of Results

Stage(s) 1, 2, 3: Fixed cost and irreversibility implications over fixed γ_{adj}.

SMM: Parameter estimation output

Policy Functions: V and i^* over z levels.

Stationary Distribution: μ density, moments, and interpretation.

2 Numerical Methods

2.1 Grid Construction

We constructed grids for capital and for productivity. We will denote said grids as K and Z respectively. As we had to accommodate the $AR(1)$ process for productivity, we took Z as a baseline grid and formed K to fit the expectations created by Z. Given Z, our main benchmark for the K specifications was eliminating (i.e. minimizing) any value maximizing capital at the grid boundaries.

2.1.1 Z Grid

The Z grid is constructed by the Rouwenhorst method which is a discretization method for approximating continuous state space Markov processes. The process allow us to create a discrete grid that captures the properties of the underlying continous $AR(1)$ process for productivity.

2.1.2 K Grid

The K grid's construction depends on Z; specifically, we use $k^*(z)$, the optimal capital choice for a given productivity level z, to determine the range of our K grid. We solve the convex cost case $\{F = 0,\ \rho_s = 1\}$ and solve for the steady state k_{ss}. The grid is then defined in the interval $[\alpha k_{ss}, \beta k_{ss}]$ where α and β parameters are chosen to ensure optimal capital choices never fall outside the grid. To achieve this, we minimize the point mass of choices at the grid boundaries through the bound scaling parameters:

$$\begin{aligned} \hat{\alpha} &= \underset{\alpha}{\operatorname{argmin}}\ R_\alpha := \frac{\sum \mathbb{1}_{\{k^*(z)<\alpha k_{ss}\}}}{N} \rightarrow \hat{\alpha} = 0.01\ (R_\alpha = 0.00\%) \\ \hat{\beta} &= \underset{\beta}{\operatorname{argmin}}\ R_\beta := \frac{\sum \mathbb{1}_{\{k^*(z)>\beta k_{ss}\}}}{N} \rightarrow \hat{\beta} = 25\ (R_\beta = 0.20\%) \end{aligned} \tag{1}$$

NOTE: We chose not to push $\hat{\beta}$ further than 25 to capture all outliers. We achieved $R_\beta = 0.2\%$ at $\beta = 25$ and tested that the next increment would happen around $\beta = 50$. At which point we decided accepting $R_\beta = 0.2\%$ as a reasonable tolerance since a marginal reduction in R_β would cost increasingly larger β specifications.

With the above result yielding a grid over $[0.01k_{ss}, 100k_{ss}]$, we observe multiple orders of magnitude between the grid bounds. Therefore, we elect to use a non-linear spacing for the K grid to emphasize the lower end while covering the upper extremes. A log-spaced grid satisfies this requirement and is constructed by defining an auxiliary grid K_{aux} with linear spacing over $[\log(\alpha k_{ss}), \log(\beta k_{ss})]$ and then exponentiating the values to obtain the final K grid:

$$\begin{aligned} K_{aux} &= \{\ln(\alpha k_{ss}), \ln(\alpha k_{ss}) + \Delta, \ln(\alpha k_{ss}) + 2\Delta, \ldots, \ln(\beta k_{ss})\} \\ K &= \{\exp(k_i)\ \forall k_i \in K_{aux}\} \end{aligned} \tag{2}$$

2.2 Interpolation

With a discrete grid, we need to either choose the nearest grid point or interpolate to evaluate the value function at non-grid points. We choose to use a weighted interpolation of the two nearest grid points for capital. Interpolation helps sanitize the kinks in value and policy functions, making them more interpretable.

Our interpolation method normalized the distance from/to the nearest grid points over the total distance between the two grid points encapsulating the optimal value. Denoting a general case where we want to interpolate a function f at a point x that lies between two grid points $\bar{x}_i$ and $\bar{x}_{i+1}$, the interpolation is given by:

$$
\begin{aligned}
f(x) &= w_i f(\bar{x}_i) + w_{i+1} f(\bar{x}_{i+1}) \\
w_i &= \frac{\bar{x}_{i+1} - x}{\bar{x}_{i+1} - \bar{x}_i} \\
w_{i+1} &= \frac{x - \bar{x}_i}{\bar{x}_{i+1} - \bar{x}_i}
\end{aligned} \tag{3}
$$

2.3 Value Function Iteration

The Value Function Iteration (VFI) process is aided by a bellman operator that uses golden-section search to find the optimal capital choice. Golden-section search refers to an iterative optimization technique for finding the extrema of a unimodal function over some interval $[a, b]$. The method works by evaluating the function at two interior points, c and d, which are determined by the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$.

Additionally, we apply a Howard improvement every 5 iterations to speed up convergence. Howard improvement begins by picking the optimal policy from the previous iteration and evaluating the value function under this policy for n periods (or to infinity by analytical definition). Afterwards, the policy is updated by finding the optimal choice under the new value function and the process is repeated until convergence. Howard improvement can significantly reduce the number of iterations in the outer VFI loop necessary for convergence, especially when the policy function stabilizes before the value function does.

2.4 MoM Estimation Setup

We use the given LRD moments as targets and match the moments from a stationary distribution (μ) computed under some parameter composition $\{\gamma_{adj}, F, \rho_s\}$. The parameters are updated by minimizing the L2 distance between target and output moments using the Nelder-Mead optimization method. However, we prefer running the optimization in an unconstrained space, so appropriate transformations are applied to the parameters prior to the optimization.

$$
\begin{aligned}
\gamma_{adj} \in \ (0, \infty) \to \gamma'_{adj} \in \mathbb{R} \quad &| \quad \gamma'_{adj} = \ln(\gamma_{adj}) \\
F \in \ [0, \infty) \to F' \in \mathbb{R} \quad &| \quad F' = \max\{10^{-12},\ \ln(F)\} \\
\rho_s \in \ [0, 1] \to \rho'_s \in \mathbb{R} \quad &| \quad \rho'_s = \text{logit}(\rho_s) = \ln\left(\frac{\rho_s}{1 - \rho_s}\right)
\end{aligned} \tag{4}
$$

With said transformations, the Nelder-Mead optimization can run unbounded over the set $\{\gamma'_{adj}, F', \rho'_s\}$ as long as we use the inverse transformations to compute moments inside the objective function.

3 Interpretation of Results

The interpretation section will cover two main parts. Initially, we will display the effects of fixed costs and irreversibility by keeping γ_{adj} constant at 2.0. Subsequently, we will estimate a complete set of parameters, including γ_{adj} and explore the outputs.

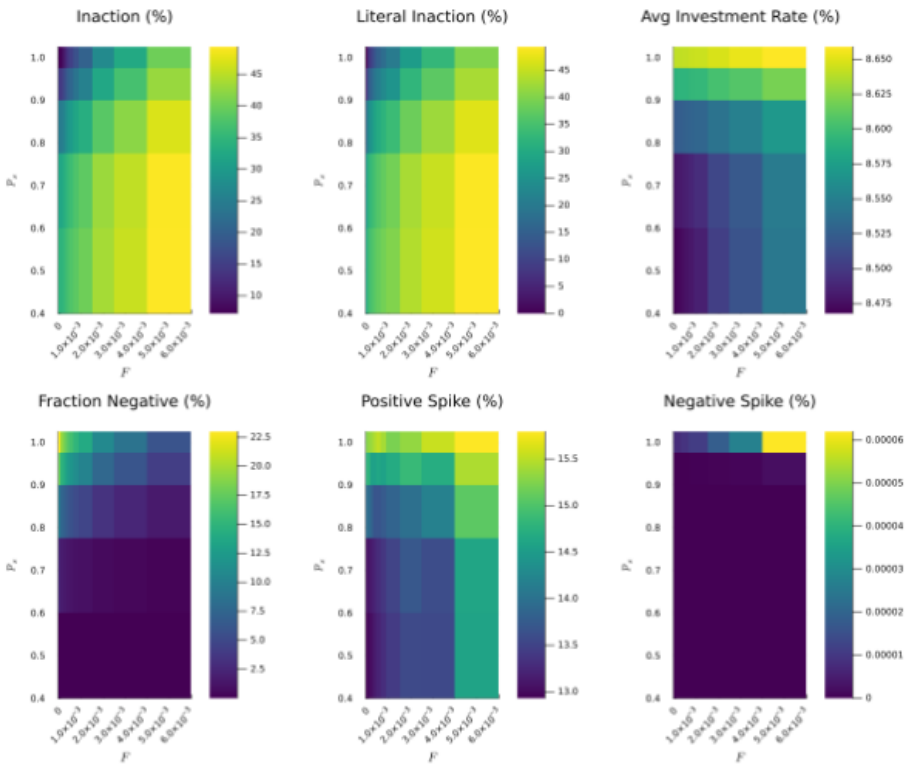

Figure 1: Moments over a grid of $\{F, \rho_s\}$ combinations.

Figure 1 computes the moments included in LRD and an additional metric "Literal Inaction". While "Inaction" includes small adjustments ($< 1\%$ of capital), "Literal Inaction" refers to the choice of inaction itself.

Intuitively, inaction should increase with both fixed costs and irreversibility, which we observe in the moments. Similarly, positive investment spikes follow an intuitive pattern, increasing with fixed costs and is relatively constant with respect to irreversibility. Notably, we observe no negative investment spikes in the data where $\rho_s \neq 1.0$ and both spikes of both signs tend to increase with fixed costs.

3.1 Case 1: Convex Costs

Case 1 depicts complete absence of fixed costs and irreversibility, which results in a convex adjustment cost function. The expected result in this case would be adjustments in all periods. With no consequences for adjusting, the firm would always choose to adjust capital to the optimal level.

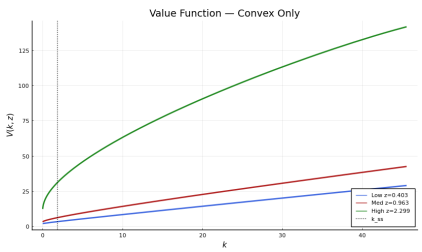

Figure 2: Value Function for z levels in Case 1

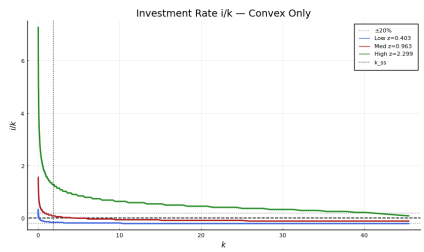

Figure 3: Investment Rates for z levels in Case 1

Figure 2 depicts the value function at varying levels of productivity z. We observe a smooth function and an increase in slope for higher z levels, which is consistent with the intuition that higher productivity should lead to higher returns on capital and thus a steeper value function. Investment rates in Figure 3 show that high productivity firms never choose to disinvest while both low and medium productivity firms monotonically stay below 0 investment rates from a relatively early point.

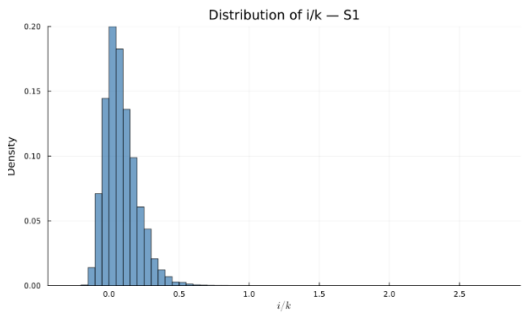

Figure 4: Histogram of Investment Rates for z levels in Case 1

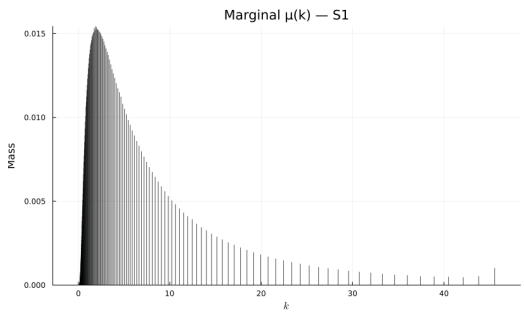

Figure 5: Marginal Capital for z levels in Case 1

The histogram depicts a t-Student like distribution of investment rates centered around 0. Notably, we see that positive adjustments are much more preferred in this parameterization. The marginal distributrion of capital also shows a intuitive pattern where higher k recieve decreaseing probability mass.

3.2 Case 2: Fixed Costs

Case 2 introduces fixed costs to the model with $F = 0.05$ while keeping reversibility intact with $\rho_s = 1.0$. As per the moment heatmap in Figure 1, we would expect an increase in inaction compared to the first case.

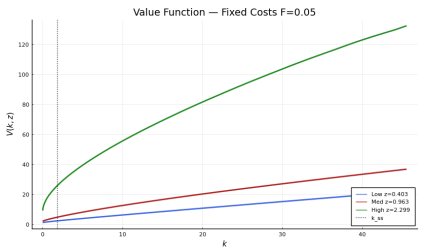

Figure 6: Value Function for z levels in Case 2

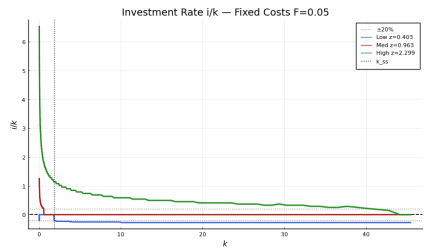

Figure 7: Investment Rates for z levels in Case 2

The value function in Figure 6 shows a similar pattern to the first case; however, we see that the computed values are slightly lower in comparison. This is perfectly sensible as the presence of added costs to the model. The investment rates in Figure 7 tell a more interesting story. Looking at the plot, we observe that low productivity firms overwhelmingly choose to disinvest while medium firms remain inactive. High productivity firms follow a relatively similar path to their first case counterparts.

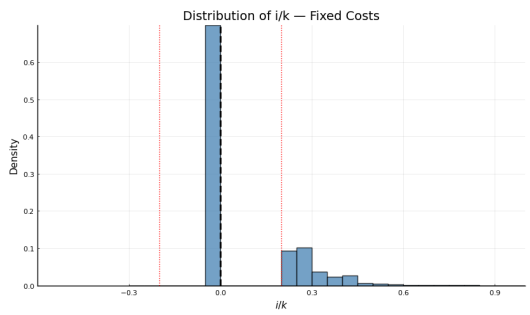

Figure 8: Histogram of Investment Rates for z levels in Case 2

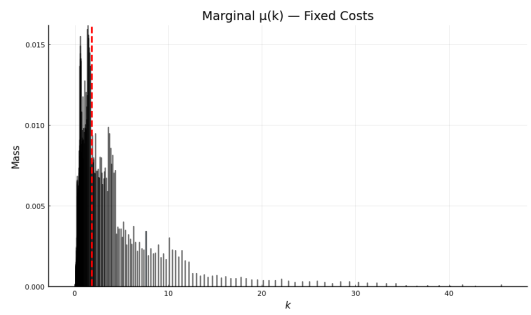

Figure 9: Marginal Capital for z levels in Case 2

While the marginal distribution of capital in Figure 9 is relatively consistent with the previous case, we see the suspected increase in inactivity with a comparatively massive peak around 0. Also, it's important to note that the bin around 0 on the negative side, including the disinvestment we

observe with low productivity firms.

3.3 Case 3: Fixed Costs + Irreversibility

The final case introduces irreversibility to the model with $\rho_s = 0.5$ and keeps the fixed costs from the previous case. Again referring to the moment heatmap in Figure 1, we expect an increase in inaction similar to the previous case and a decrease in negative investment.

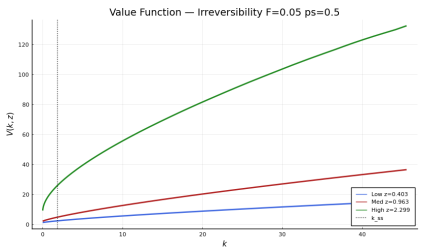

Figure 10: Value Function for z levels in Case 3

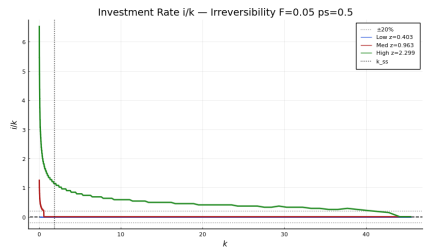

Figure 11: Investment Rates for z levels in Case 3

We see that the value function in Figure 10 is near undifferentiable from that of case 2. We can reason that this is likely caused by inactivity being already very high in case 2 and the addition of irreversibility does not change the value function much. The investment rates in Figure 11 shows a similar picture with a key difference compared to case 2. We observe that no class of firms choose to disinvest, which is consistent with the intuition that irreversibility should discourage negative adjustments.

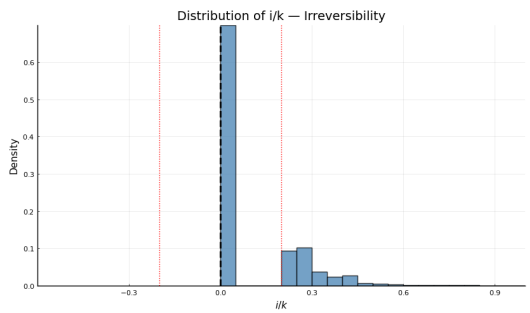

Figure 12: Histogram of Investment Rates for z levels in Case 3

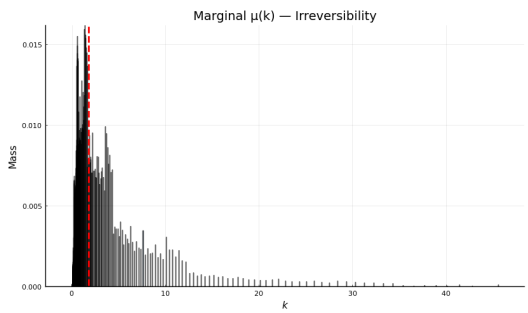

Figure 13: Marginal Capital for z levels in Case 3

The histogram of investment rates in Figure 12 is basically identical to case 2 with again a single

key difference. Due to the lack of disinvestment, we see that the spike on 0 is positioned towards the positive side of the distribution. However, the marginal distribution of capital shows no identifiable difference between the two cases.

3.4 MoM Estimation of Model Parameters

After analyzing the cases, we move onto estimating the parameters using MoM against the given data. We apply estimation via moment computations from the stationary distributions and do not use Monte-Carlo or otherwise simulate data.

Applying said methodology, we obtain the following parameter estimates:

$$\hat{F} = 0.00001 \tag{5}$$

$$\hat{\rho}_s = 0.9978 \tag{6}$$

$$\hat{\gamma}_{adj} = 2.9703 \tag{7}$$

Figure 14: Estimated Parameters

The moments obtained from the estimated parameters are compared to the cases as follows:

Moment	LRD Target	Case 1	Case 2	Case 3	Final Calibration
Avg. Investment Rate (%)	12.2	8.63	8.96	8.96	8.42
Inaction Rate (%)*	8.1	11.31	69.86	69.9	9.32
Fraction Negative(%)	10.4	21.88	0.02	0.0	16.14
Positive Spike Rate(%)*	18.0	15.6	30.12	30.1	9.52
Negative Spike Rate(%)*	1.4	0.0	0.02	0.0	0.0

* Moments that are targeted in the objective function

Table 1: Estimated moments and moments from cases

We see that the estimates are rather inaccurate in matching the target moments. This could partly be attributed to our difficulty in getting average investment rates to adjust over 9% and the fact that we could not estimate non-zero negative spikes even when targeting it by itself. However, we picked our target moments to each correspond to (and contribute information related to) a specific parameter. Specifically, we chose inaction rates as a target to capture the fixed costs, while positive spike rates were chosen to capture the positive effect of γ_{adj} on investment rates. Similarly, we chose negative spike rates to capture the negative effect of the presence of irreversibility.